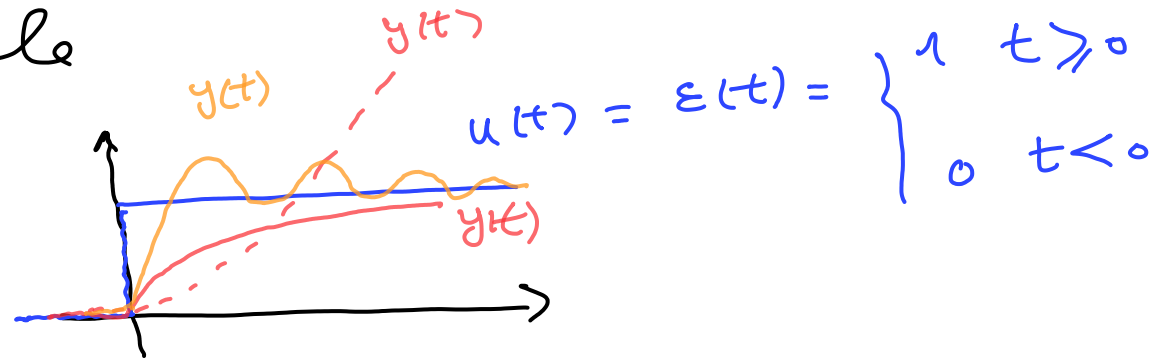
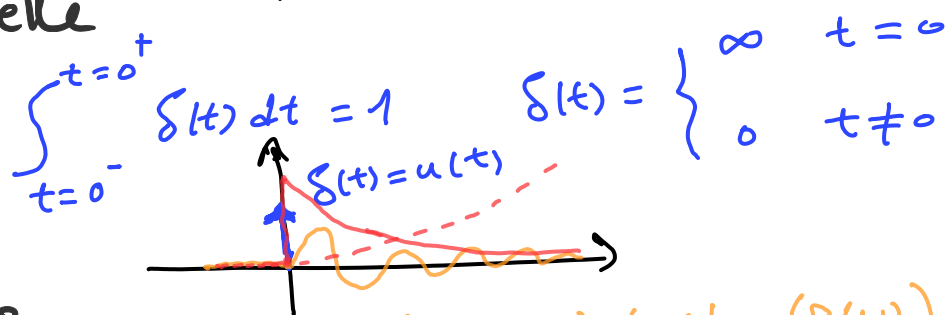
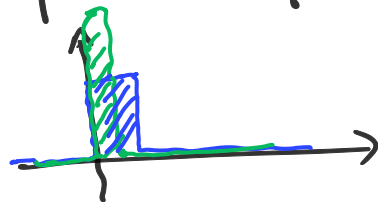


① Approche expérimentale

- réponse indicielle

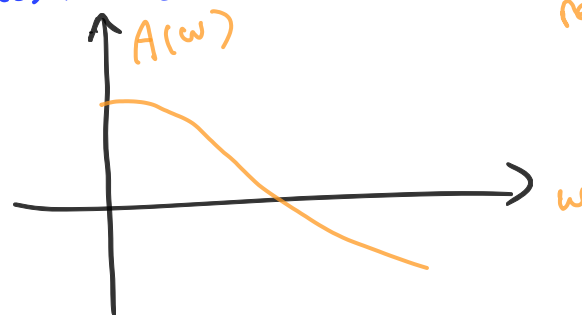


- réponse impulsionnelle

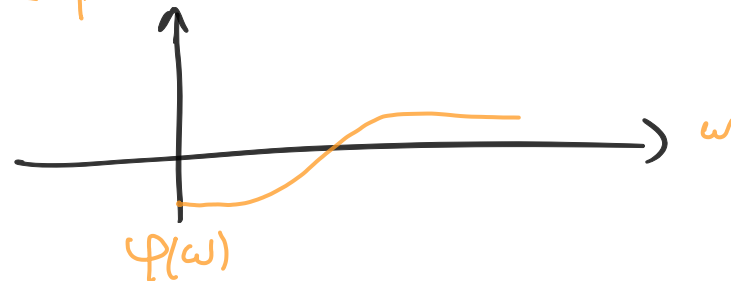


- réponse fréquentielle

$u(t) = \sin(\omega t)$



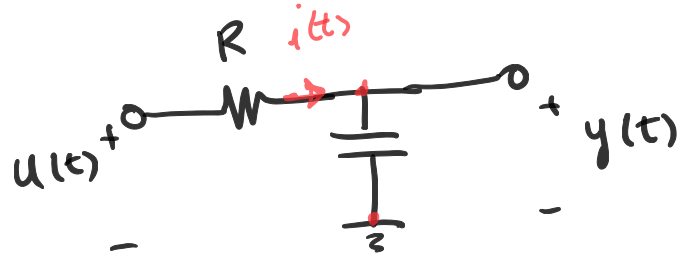
$y(t) = A(\omega) \sin(\omega t + \varphi(\omega))$
régime permanente



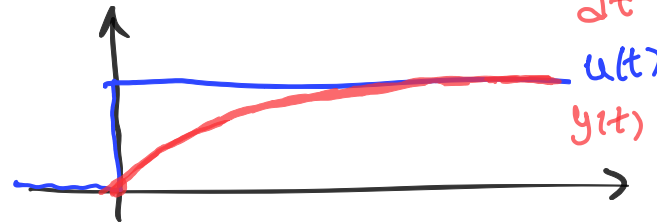
② Approche (modélisation) théorique

2.1 Equation différentielle

exemple;
RC



Rep. indicelle
 $u(t) = \varepsilon(t)$



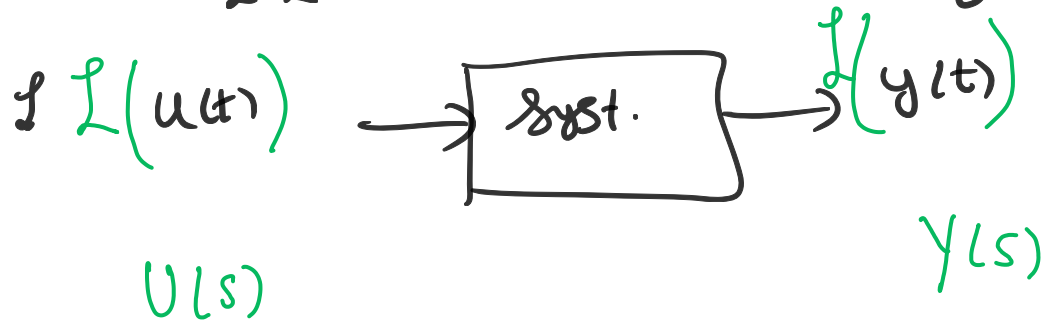
$i(t)$ variable intermédiaire
 $= C \cdot \frac{dy}{dt}$

$$a_n \frac{d^n y(t)}{dt^n} + a_{n-1} \frac{d^{n-1} y(t)}{dt^{n-1}} + \dots + a_1 \frac{dy}{dt} + a_0 y(t) = b_m \frac{d^m u(t)}{dt^m} + \dots + b_1 \frac{du}{dt} + b_0 u(t)$$

$$u(t) - y(t) = R \cdot i(t) \Rightarrow$$

$$u(t) = y(t) + R \cdot C \cdot \frac{dy}{dt}$$

2.2 Fonction de transfert (Transfer Function, tf)



$$U(s) = Y(s) + R \cdot C \cdot s \cdot Y(s)$$

$$\hookrightarrow \frac{Y(s)}{U(s)} = \frac{1}{1 + R \cdot C \cdot s}$$

$$L(u(t)) = \int_{t=-\infty}^t u(t) \cdot e^{-st} dt$$

Caus Regu 0

$$G(s) = \frac{Y(s)}{U(s)}$$

fonction transfert

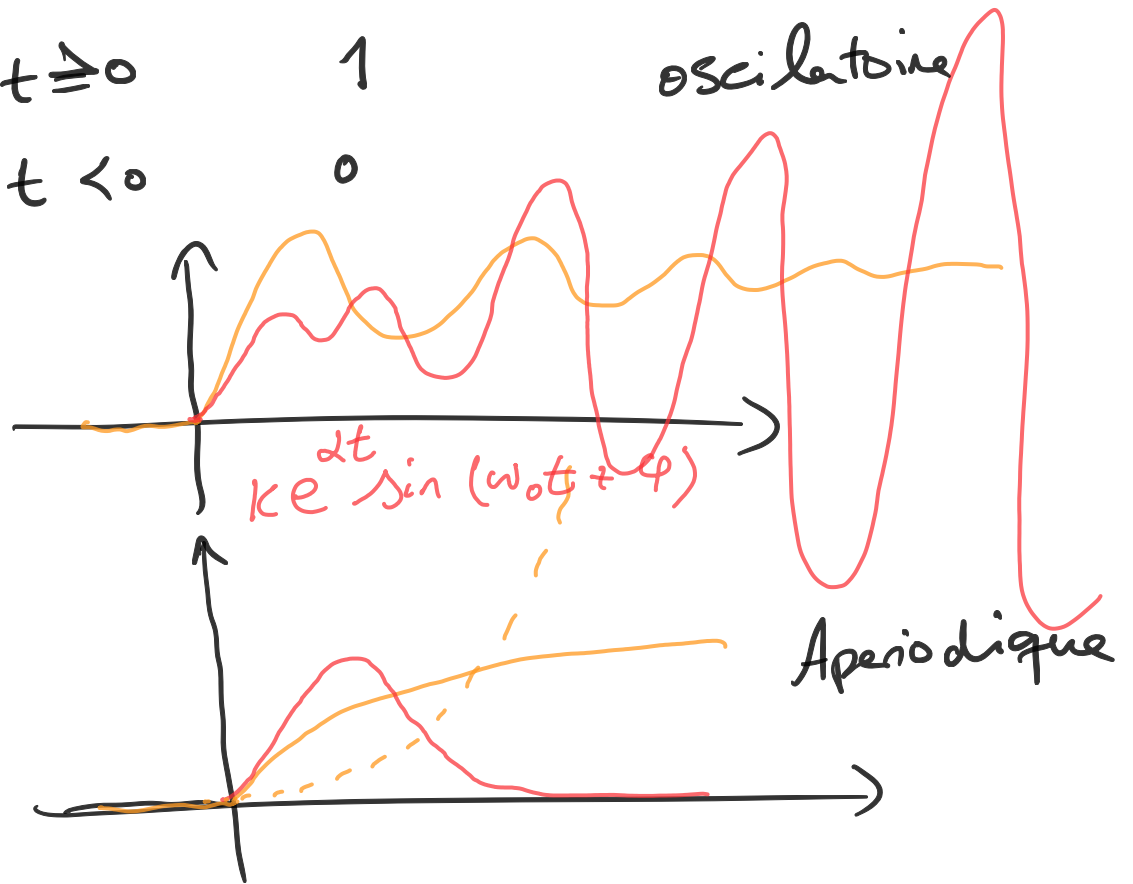
$$L^{-1} Y(s) = G(s) \cdot U(s)$$

$$\hookrightarrow y(t) = L^{-1}(G(s) \cdot U(s))$$

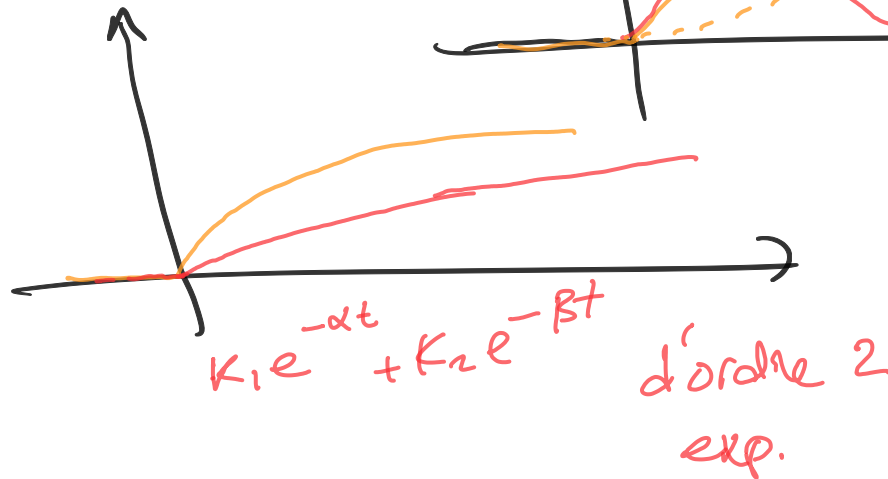
Reponse Indicielle

$$u(t) = \varepsilon(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

{ Aperiodique
 oscillatoire



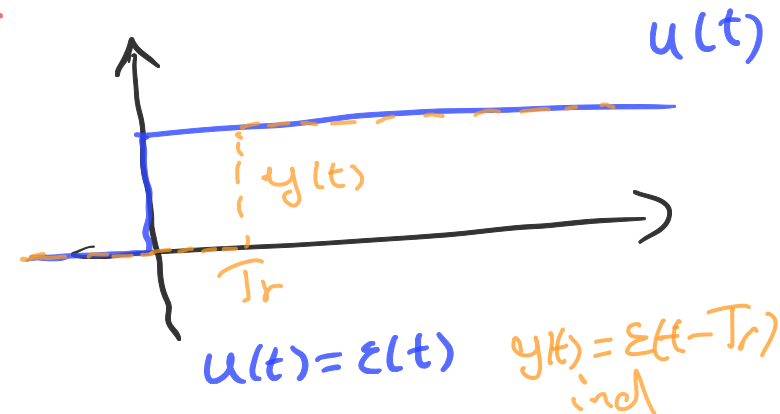
d'ordre 1
 exp. $e^{-\alpha t}$



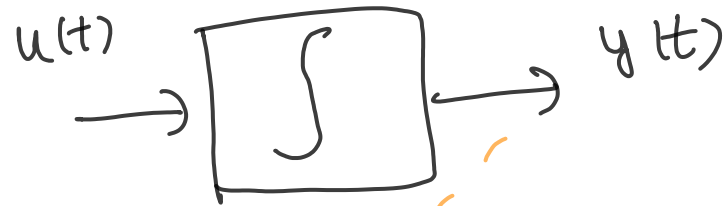
retard pur

$$y(t) = u(t - T_r)$$

↳ retard



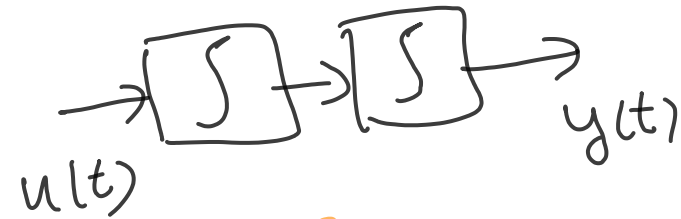
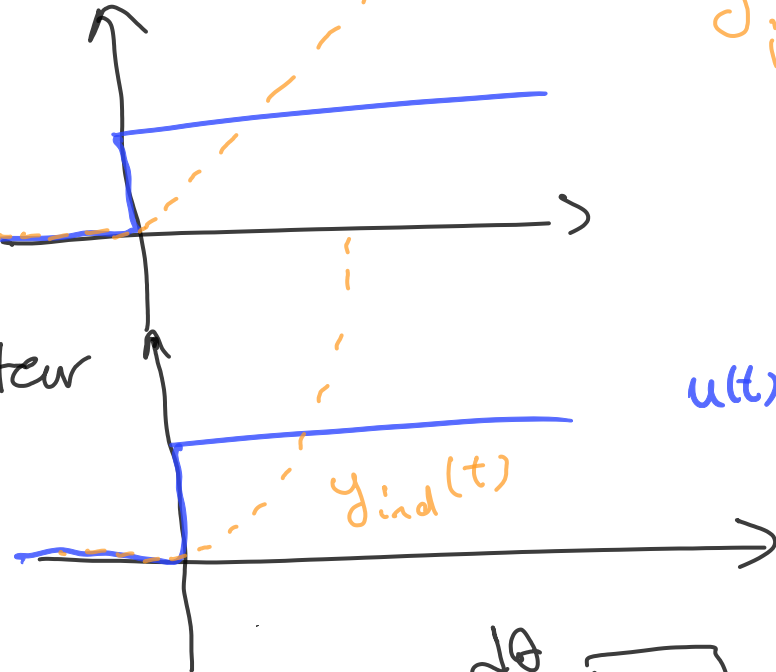
Integrator.



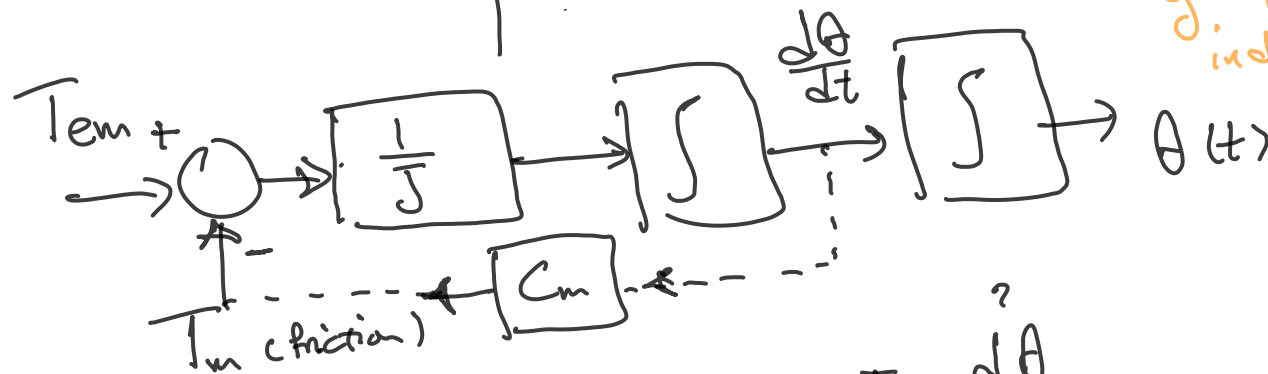
$$u(t) = \varepsilon(t)$$

$$y_{ind}(t) = \int_{\tau=0}^{\tau=t} \varepsilon(\tau) d\tau = t \cdot \varepsilon(t)$$

double Integrator



$$y_{ind}(t) = t^2 \cdot \varepsilon(t)$$



$$T_{em} - T_{friction} = J \cdot \frac{d\omega}{dt} = J \cdot \frac{d^2\theta}{dt^2}$$

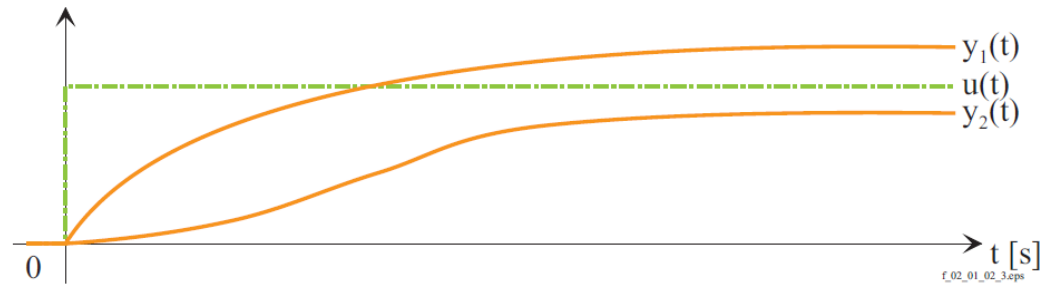


FIGURE 2.6 – Réponses indicielles d'un système d'ordre 1 ($y_1(t)$) et d'un système d'ordre élevé ($y_2(t)$).

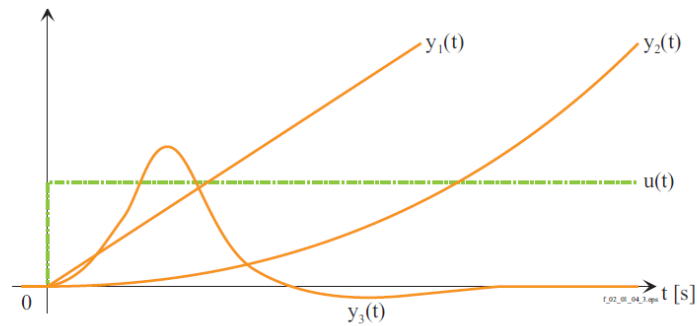


FIGURE 2.10 – Réponses indicielles d'un intégrateur pur ($y_1(t)$), de 2 intégrateurs purs ($y_2(t)$) et d'un système à comportement (entre autre) dérivateur ($y_3(t)$)

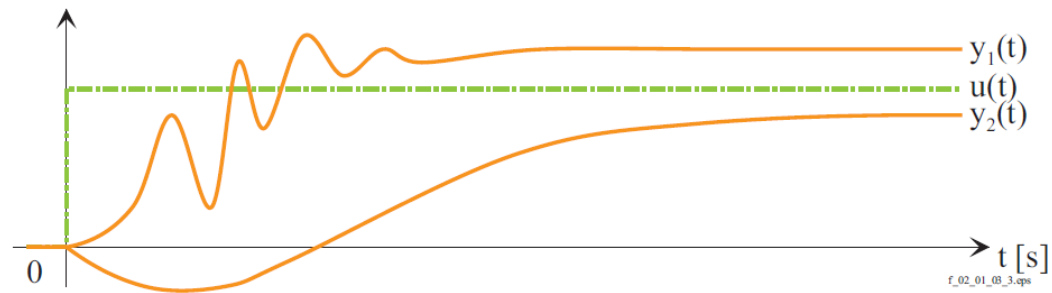
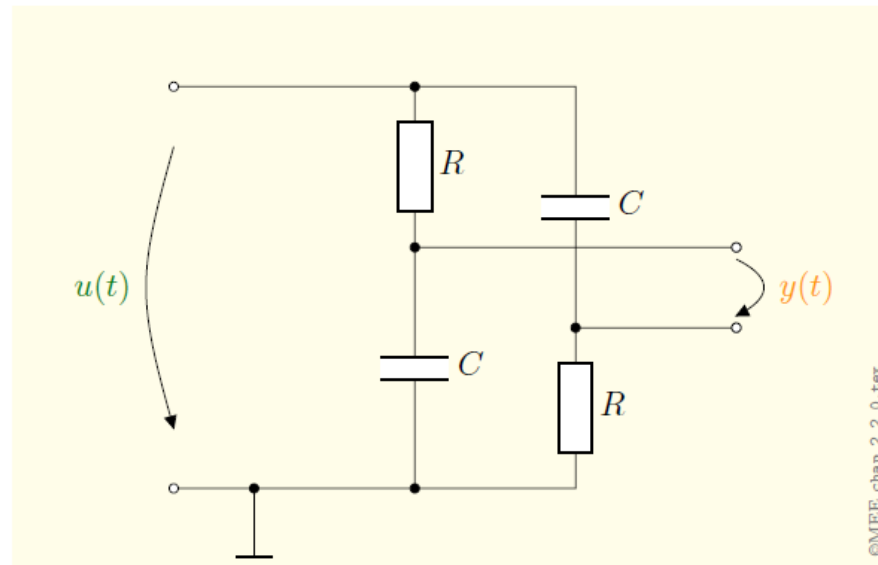
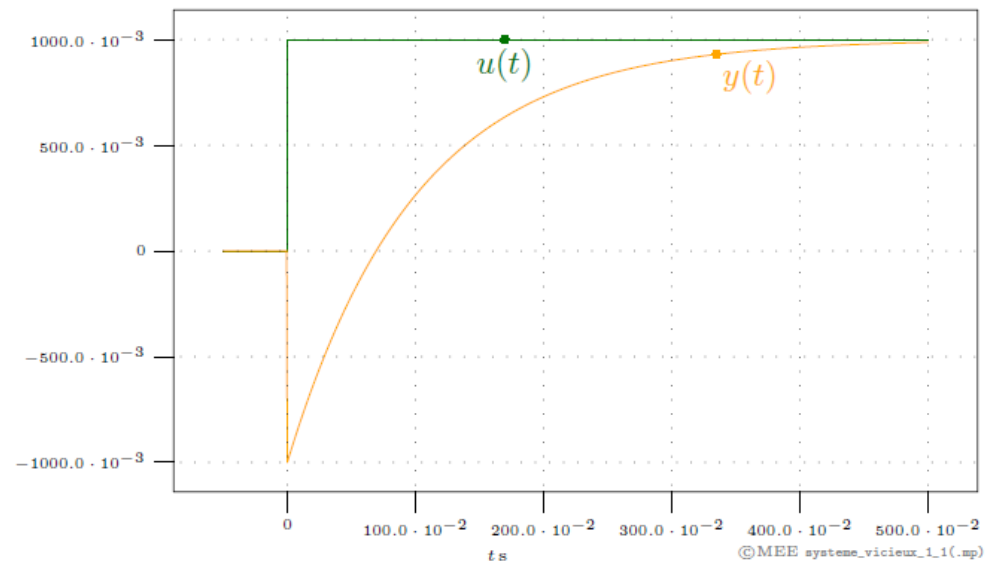


FIGURE 2.7 – Réponses indicielles d'un système oscillant ($y_1(t)$) et d'un système à déphasage non-minimal ou « vicieux » ($y_2(t)$) .

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(a) Schéma technologique d'un système à déphasage non-minimal ou « vicieux ». Voir sa réponse indicielle sur la figure 2.9b .



(b) Réponse indicielle du système à déphasage non-minimal de la figure 2.9a.

FIGURE 2.9 – Système « vicieux »

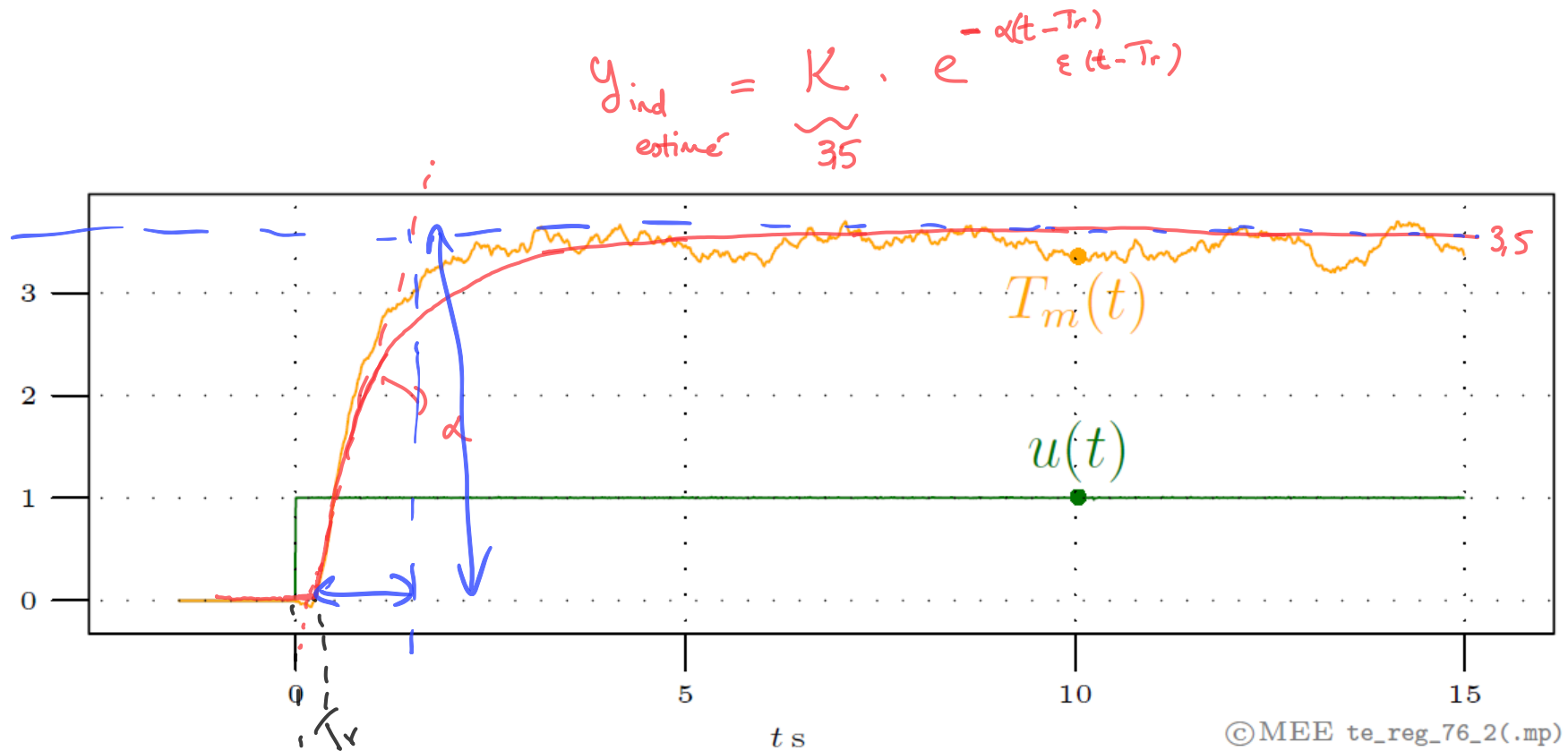
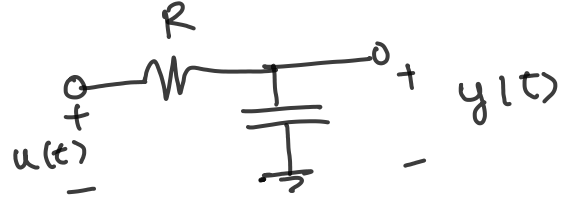
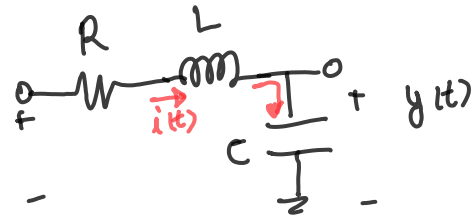


FIGURE 2.5 – Réponse indicielle d'un système à retard pur : canal aérothermique (« foehn ») du laboratoire d'automatique de la [HEIG-VD](#). Le signal d'entrée est un saut de tension aux bornes du corps de chauffe, celui de sortie est la température mesurée. On observe un retard pur de l'ordre de $T_r = 200$ ms .

Equation differentielle



$$y(t) + RC \frac{dy}{dt} = u(t)$$



$$u(t) = y(t) + R \cdot i(t) + L \frac{di}{dt}$$

$$i(t) = C \cdot \frac{dy}{dt} \Rightarrow u(t) = y(t) + R \cdot C \frac{dy}{dt} + LC \cdot \frac{d^2 y}{dt^2}$$

$$u(t) = \varepsilon(t) \Rightarrow LC \frac{d^2 y}{dt^2} + RC \frac{dy}{dt} + y(t) = \varepsilon(t)$$

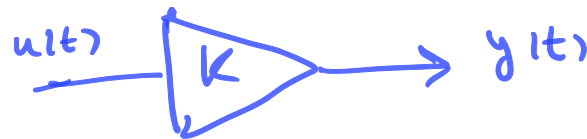
$$LC \cdot s^2 + RC \cdot s + 1 = 0$$

$$s_1 \text{ et } s_2 = \alpha \pm j\omega_0$$

Schema fonctionnelle

pres. d'une equation differentielle par les blocs generique

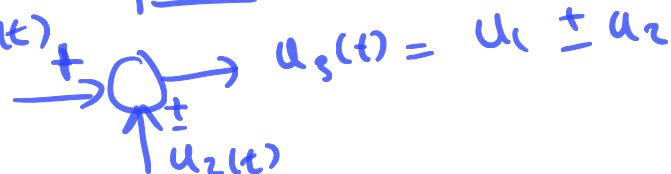
1 - gain



2 - integrateur

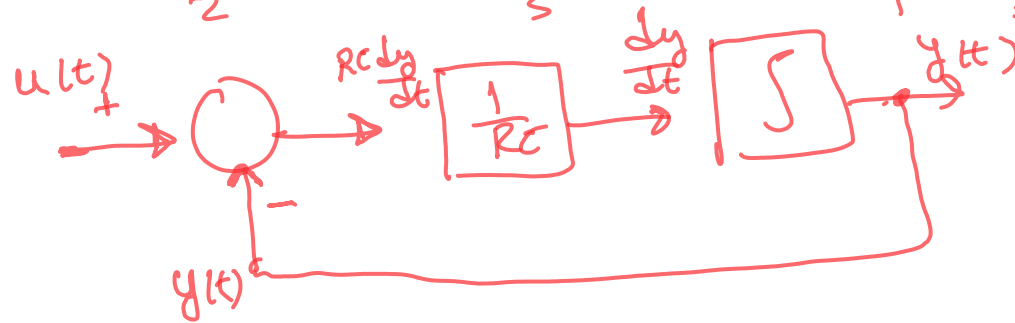


3 - Comparateur
additionner



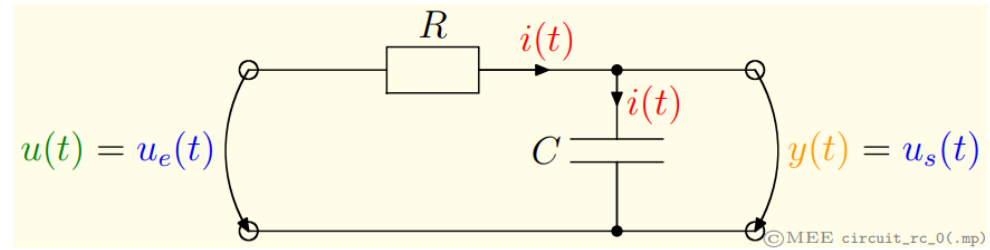
$$RC \frac{dy(t)}{dt} + y(t) = u(t)$$

1
2
3

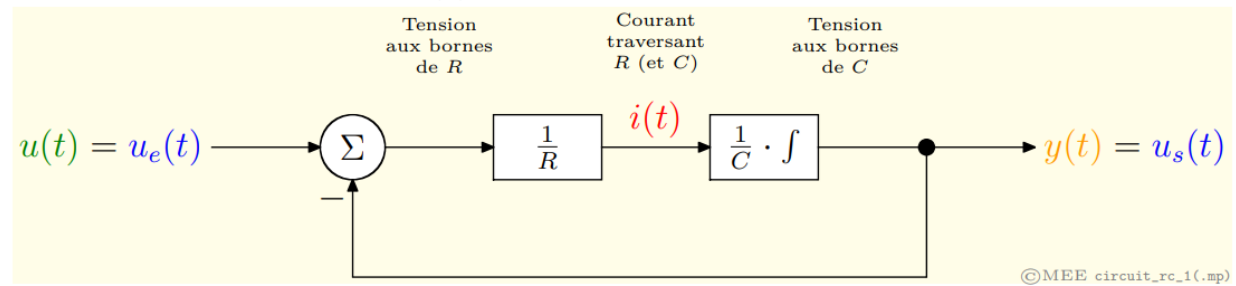


"1" + "2" = "3"

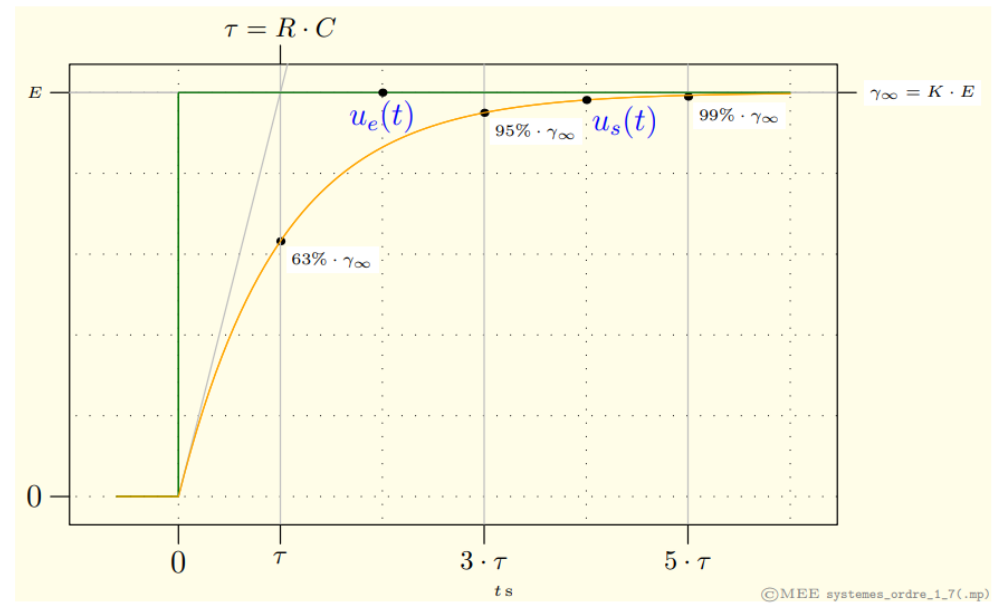
"1" = "3" - "2"



(a) Schéma technologique .



(b) Schéma fonctionnel détaillé.



(c) Réponse indicielle.

FIGURE 2.15 – Filtre passe-bas de type circuit RC.

